

Phase 2 Test Case Report

7 / 7 automated tests passed | Combined 20 / 20 | 15 September 2026

AI document analysis tests: authentication, ownership, mocked Sarvam pipeline, deterministic risk engine, evidence, and recommendations. Suite: backend/tests/test_phase2.py. pytest does not call the live Sarvam API.

1. Summary

Phase 2 passed	7
Phase 1 still passing	13 (same pytest run)
Combined	20 / 20
Last duration	about 21.5 seconds
Harness	pytest 9.1.1 + FastAPI TestClient
Sarvam in pytest	Mocked
Verdict	Phase 2 automated tests passed.

2. Automated cases

TC-21 Analyze requires authentication

Function: test_analyze_requires_authentication

Steps: POST analyze with no JWT.

Expected: 401. Actual: 401. Status: Pass

TC-22 Analyze uploaded PDF (mocked Sale Deed)

Function: test_analyze_uploaded_pdf

Steps: Upload PDF. Mock Kannada OCR and English translation. POST analyze. GET analysis.

Expected: 200 SALE_DEED COMPLETE. Litigation DETECTED/HIGH. Mortgage NOT_VERIFIED. Approval NOT_VERIFIED. Recommendations present.

Actual: As expected. Status: Pass

TC-23 Cannot analyze another user's document

Function: test_user_cannot_analyze_another_users_document

Steps: Owner uploads. Second user POSTs analyze.

Expected: 403. Actual: 403. Status: Pass

TC-24 Missing document returns 404

Function: test_missing_document_returns_404

Steps: Analyze a random document UUID on a valid case.

Expected: 404. Actual: 404. Status: Pass

TC-25 Litigation detected from explicit evidence

Function: test_litigation_detected_from_explicit_evidence

Steps: AI JSON has stay_order=null. Source text has O.S. No. 234/2019, pending, stay, restriction. GET evidence.

Expected: DETECTED, case 234/2019, stay true, evidence list not empty.

Actual: As expected. Status: Pass

TC-26 Absence is not NO_ISSUE_FOUND

Function: test_empty_evidence_does_not_produce_no_issue_found

Steps: Analyze a clean Sale Deed with no court language.

Expected: Litigation, Mortgage, Approval = NOT_VERIFIED. Not NO_ISSUE_FOUND.

Actual: As expected. Status: Pass

TC-27 Recommendation engine

Function: test_recommendation_engine_returns_missing_documents

Steps: Analyze mocked litigation Sale Deed. GET /recommendations.

Expected: Includes ENCUMBRANCE_CERTIFICATE, COURT_ORDER, BUILDING_PLAN_APPROVAL.

Actual: As expected. Status: Pass

3. Live / manual notes

TC-20 Analyze Document UI uses the real analyze API. TC-29 Live Kannada PDF (SaleDeed_01_Litigation_Kannada.pdf): first run failed because Sarvam OCR download is a ZIP. That is fixed. Click Run analysis again with SARVAM_API_KEY set.

4. Re-run

cd backend

pytest tests/test_phase2.py -v

pytest tests/test_phase1.py tests/test_phase2.py -v

5. Out of scope

Blockchain, final due-diligence PDF, cross-document comparison, real payments, and real Google/X/phone login are not Phase 2 tests. Live Sarvam is not used inside pytest.